Reading and Sketching Slope Fields
Answer Key

AP Calculus AB/BC Differential Equations and Numerical Solution

Quick Answers

1. 1, 3, -3, -2
2. 2, -1, -2, 2
3. 2, -1, 0
4. x^2, 2, 1
5. 0, 3, x
6. $\frac{x^3}{3} - x$, 1, 1.67
7. 2, 0, -2, $-x$
8. 3, 1, -1

Curriculum

Level: AP Calculus AB/BC Differential Equations and Numerical Solution
FUN-7 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 30 tagged checks.

Common wrong answers

1. *If they got -1: subtracted in the wrong order, computing y minus x*
6. *If they got 0.67: forgot the initial condition and evaluated the general antiderivative with no constant*

1. $\dfrac{dy}{dx} = x - y$ — slope table.

Step 1. The right-hand side is the slope. Substitute each point's own coordinates, in the order x then y; nothing is solved, only evaluated.
Step 2. At $(1, 0)$: $1 - 0 = \boxed{1}$
At $(2, -1)$: $2 - (-1) = \boxed{3}$
At $(-1, 2)$: $-1 - 2 = \boxed{-3}$
At $(0, 2)$: $0 - 2 = \boxed{-2}$
Step 3 (the drawing). Segments of slope 1, 3, -3 and -2 at those four points. Draw them short — a slope field segment shows direction, not length — and steeper for the larger magnitudes.
Common wrong answer: -1 at the first point — the subtraction was done in the wrong order, as $y - x$.

2. $\dfrac{dy}{dx} = 2 - y$**.**
Step 1. There is no x on the right-hand side, so the slope depends only on the height. Every segment in a horizontal row is parallel.
Step 2. At height $y = 0$: $2 - 0 = \boxed{2}$
At height $y = 3$: $2 - 3 = \boxed{-1}$
At height $y = 4$: $2 - 4 = \boxed{-2}$
Step 3. Horizontal segments need slope zero: $2 - y = 0$, so the constant solution is the line $\boxed{y = 2}$
Step 4 (the drawing). A row of horizontal segments along that line, a row of gentle upward segments along the bottom, and rows tilting downward more and more steeply above it.
Reading it: solutions below the line rise toward it and solutions above it fall toward it, so the constant solution attracts everything.

3. Which equation made this field?
Step 1. Every segment in a given vertical column of the printed field has the same slope, and the slope changes as you move left or right. That is the fingerprint of a right-hand side that contains x and no y.
Step 2. In the column $x = 4$ the segments rise one unit for every two across, a slope of $\boxed{2}$; in the column $x = -2$ they fall half a unit per unit across, a slope of $\boxed{-1}$. Both match $\dfrac{x}{2}$ exactly.
Step 3. Setting $\dfrac{x}{2} = 0$ gives $\boxed{x = 0}$, so the horizontal segments lie along the whole y-axis — which is exactly what the picture shows.
Step 4 (the justification — manual review). *Full credit says:* the field is $\dfrac{dy}{dx} = \dfrac{x}{2}$. It cannot be $\dfrac{dy}{dx} = \dfrac{y}{2}$, because that field would be constant along each horizontal row and its horizontal segments would run along the x-axis, not the y-axis. It cannot be $\dfrac{dy}{dx} = x - y$, because that field changes as you move straight up a column and its horizontal segments lie on a slanted line.

4. $\dfrac{dy}{dx} = 2x$ **through** $(1, 2)$**.**
Step 1. The slope depends only on x, so each vertical column carries one repeated segment: steeply falling on the far left, horizontal along the y-axis, steeply rising on the far right.
Step 2. Antidifferentiate: the general antiderivative of $2x$ is $\boxed{x^2}$, so every solution has the form $y = x^2 + C$.
Step 3. Fix C from the given point: $2 = 1^2 + C$, so $C = 1$ and $y = x^2 + 1$. Check it at the given point: $y(1) = \boxed{2}$ ✓
Step 4. At the bottom of the curve, $y(0) = \boxed{1}$
Step 5 (the sketch). An upward parabola with its lowest point on the y-axis one unit above the origin, riding tangent to the field's segments the whole way. The family $y = x^2 + C$ is just this curve slid up and down — and that is why the field looks the same in every horizontal row.

5. The finished field for $\dfrac{dy}{dx} = x - y$.

Step 1. At $(2,2)$ the two coordinates are equal, so the field's segment is horizontal: slope $2 - 2 = \boxed{0}$

Step 2. At $(3,0)$: $3 - 0 = \boxed{3}$, a steeply rising segment — and the picture agrees.

Step 3. Horizontal segments need $x - y = 0$, that is $\boxed{y = x}$ This is the diagonal line of flat segments running across the field.

Step 4 (the description — manual review). *Full credit says:* the curve through $(0,3)$ starts steeply downward (slope -3 there), flattens as it drops toward the diagonal, and from then on runs just below the line $y = x - 1$, climbing with it. For large x it is indistinguishable from that line: the field pushes every solution toward the same slanted asymptote.

6. $\dfrac{dy}{dx} = x^2 - 1$ through $(0,1)$.

Step 1. The right-hand side is zero when $x^2 = 1$, so the field has two whole columns of horizontal segments. Those columns are where the solution curve turns.

Step 2. Antidifferentiate: $\boxed{\dfrac{x^3}{3} - x}$ is the general antiderivative, so $y = \dfrac{x^3}{3} - x + C$.

Step 3. Fix C from the given point: $1 = 0 - 0 + C$, so $C = 1$ and $y = \dfrac{x^3}{3} - x + 1$. Check: $y(0) = \boxed{1}$ ✓

Step 4. At the right edge: $y(2) = \dfrac{8}{3} - 2 + 1 = \dfrac{5}{3}$, which to two decimal places is $\boxed{1.67}$

Step 5 (the sketch). The curve rises to a local maximum in the left-hand horizontal column, falls to a local minimum in the right-hand one, then rises again — exactly the shape the two flat columns force, and it passes back up through about 1.67 at the right edge.

Common wrong answer: 0.67 — the antiderivative was evaluated with no constant, so the initial condition was never used.

7. $\dfrac{dy}{dx} = x + y$.

Step 1. Substitute each point. At $(1,1)$: $1 + 1 = \boxed{2}$

At $(2,-2)$: $2 + (-2) = \boxed{0}$

At $(-3,1)$: $-3 + 1 = \boxed{-2}$

Step 2. Horizontal segments need $x + y = 0$, so they sit along $\boxed{y = -x}$, the falling diagonal through the origin.

Step 3 (the drawing). Around the origin the nine segments run from slope -2 at the top left, through slope 0 all along the falling diagonal, to slope 2 at the bottom right.

Step 4. The steepest segment on the grid is at the top right corner, where x and y are both as large as the grid allows and the sum $x + y$ is therefore the largest slope drawn. The bottom left corner is equally steep downward — the field is symmetric through the origin.

8. $\dfrac{dy}{dx} = \dfrac{y-3}{2}$ **— equilibrium and long-run behaviour.**

Step 1. There is no x on the right, so the field is constant along every horizontal row and the whole picture is decided by height.

Step 2. Setting $\dfrac{y-3}{2} = 0$ gives the equilibrium solution $\boxed{y=3}$: a horizontal line of horizontal segments, and a genuine constant solution of the differential equation.

Step 3. At height $y = 5$ the slope is $\dfrac{5-3}{2} = \boxed{1}$, positive — above the line the field points upward, away.

At height $y = 1$ the slope is $\dfrac{1-3}{2} = \boxed{-1}$, negative — below the line the field points downward, also away.

Step 4 (the sketches and the explanation — manual review). *Full credit says:* the curve through $(0, 5)$ starts with slope 1 and gets steeper as it climbs, running away upward; the curve through $(0, 1)$ starts with slope -1 and steepens as it falls. Neither can reach the equilibrium line, because approaching it means the slopes shrink toward zero while pointing the wrong way — the segments near the line push away from it, not toward it. Equivalently, the constant function is itself a solution through every point of that line, and two different solution curves cannot pass through the same point.

Contrast with problem 2: there the arrows pointed *toward* the constant solution. Same kind of picture, opposite long-run story — the sign of the coefficient on y is what decides it.